

The data we chose:

Algorithms and data structures developer:

Members:

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Report statistics:

Metric	Result
Indexed Documents	55,086
Unique Tokens	1,224,781
Index Size on Disk	1,503,038.62 KB ($\approx$ 1.43 GB)

Description of how we build the index:

- The indexer reads every JSON file under DEV/ and treats its content field as the raw HTML of one document. BeautifulSoup (html.parser) is used to strip <script> and <style> blocks and pull out the visible text.
- For tokenization, the text is scanned character-by-character: runs of ASCII letters and digits are accumulated into a token (lowercased), and anything else acts as a separator. Stemming and stop-word filtering are not applied for M1.
- For each document, raw term frequencies are counted, then one posting is appended to the index for every token that appeared. A posting is stored as: {"doc_id": <path relative to DEV/>, "tf": <count in document>}
- The full index is held in an in-memory defaultdict(list) while the corpus is processed, and written to inverted_index.json at the end. Document count, unique token count, and the final file size are recorded in index_report.json.